Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 40)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 1 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

0 1 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

0 1 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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0 2 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 2 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

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0 2 . 2 Explain the changes shown in Table 1.

[5 marks]

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0 2 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

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0 2 . 4 Describe the early stages of a lithosere.

[1 mark]

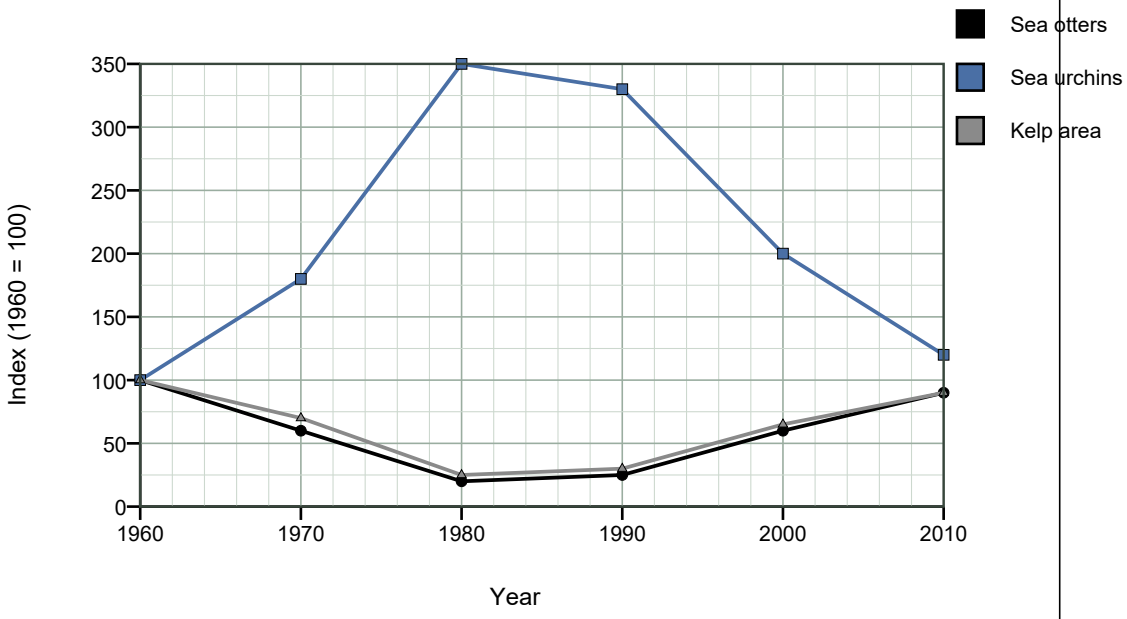
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0 3 Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 3 . 1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

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0 3 . 2 Explain the relationships shown in Figure 1.

[3 marks]

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0 3 . 3 Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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0 3 . 4 Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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0 4 Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km^2	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km^2	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km^2	under 250	over 50% in 10 years or 3 generations

0 4 . 1 State **three** roles of the IUCN.

[3 marks]

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0 4 . 2 A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km^2 of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 4 . 3 Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 4 . 4 Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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0 5 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 5 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 5 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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0 6 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 6 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 6 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 7 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems
Energy source		
Use of materials		
Nature of wastes		
Temperature of processes		

Do not write outside the box

0 7 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

0 7 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

0 7 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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0 8 . 1 Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Square-mesh escape panels .....

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Minimum and maximum catchable sizes .....

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No-take zones .....

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Closed seasons .....

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0 8 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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0 9 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

0 9 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

0 9 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

0 9 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

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1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

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1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

11 . 1 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

11 . 2 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS